Sugarcane Irrigation Advisory: Farm Overview

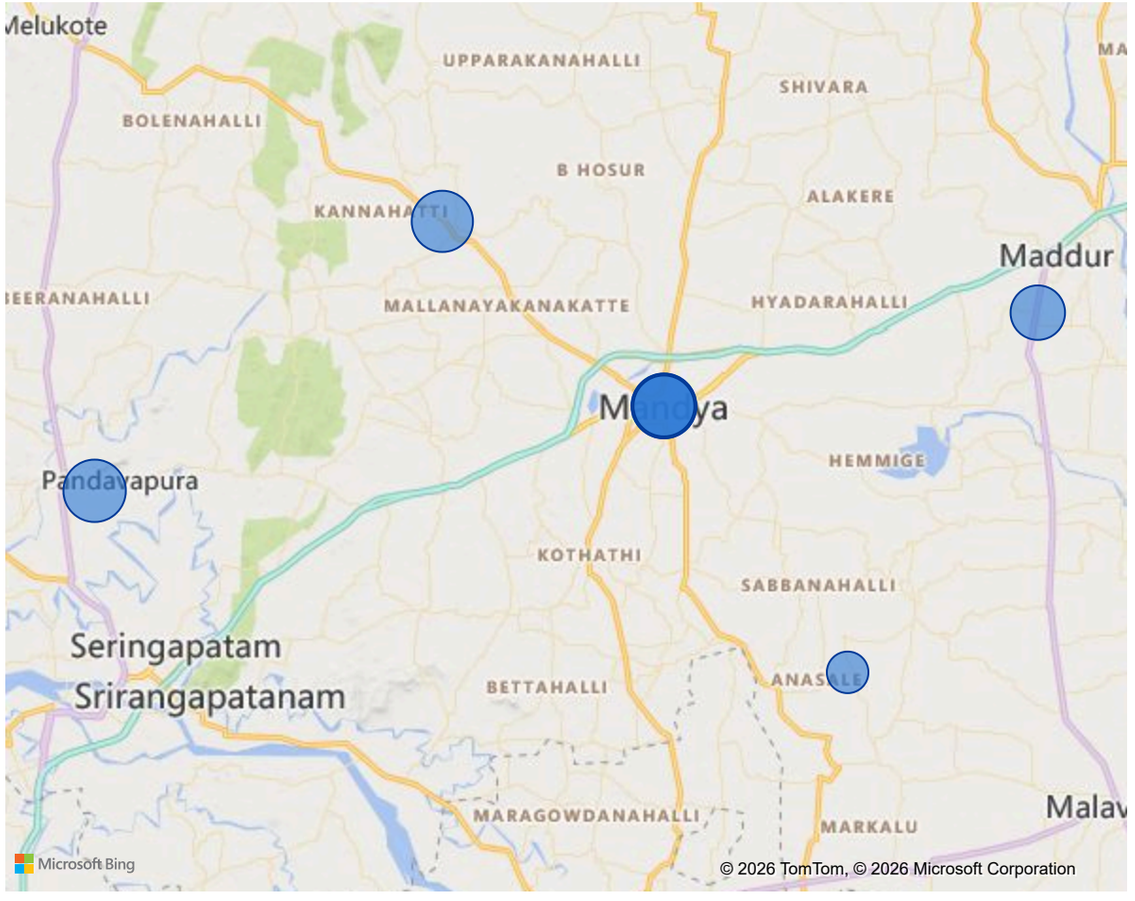
1,000 geofenced sugarcane plots in Mandya district, Karnataka (satellite + weather + soil data)

Taluk

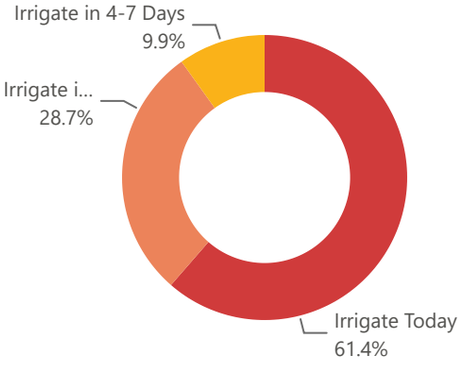
All

Plots monitored	Plots to irrigate today	Water required today	Avg water stress score (0-100)	Water saved vs farmer practice
1,000	614	3,68,680 m³	56.8	49.9%

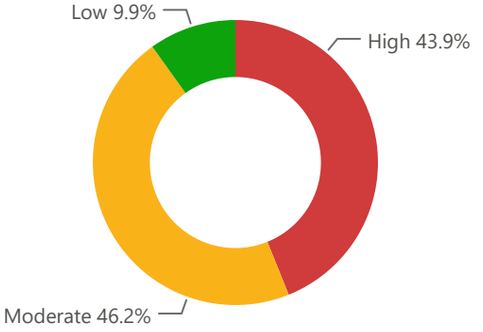
Villages: bubble size = plots to irrigate today



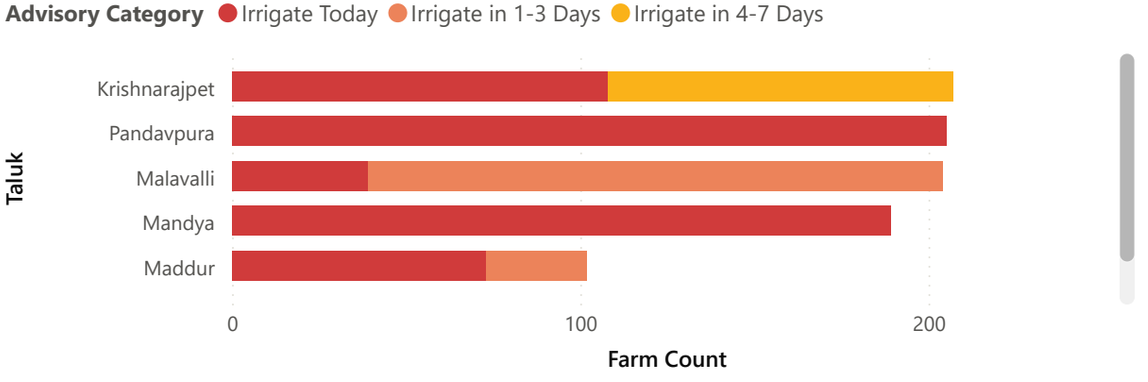
Plots by irrigation advisory



Plots by water stress level



Advisory mix by taluk



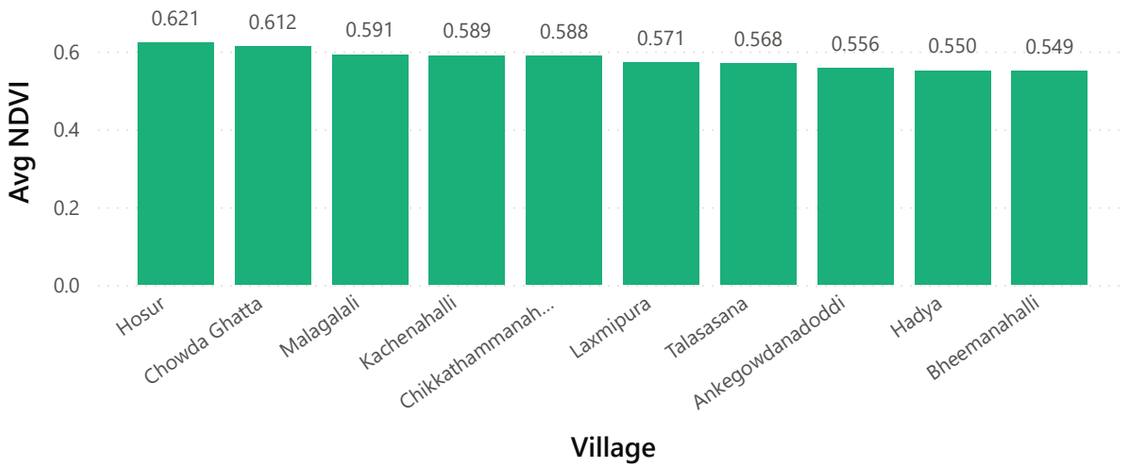
Soil Moisture and Crop Health

Hover a village bar for its tooltip. Right-click a plot in the table on page 4 to drill through.

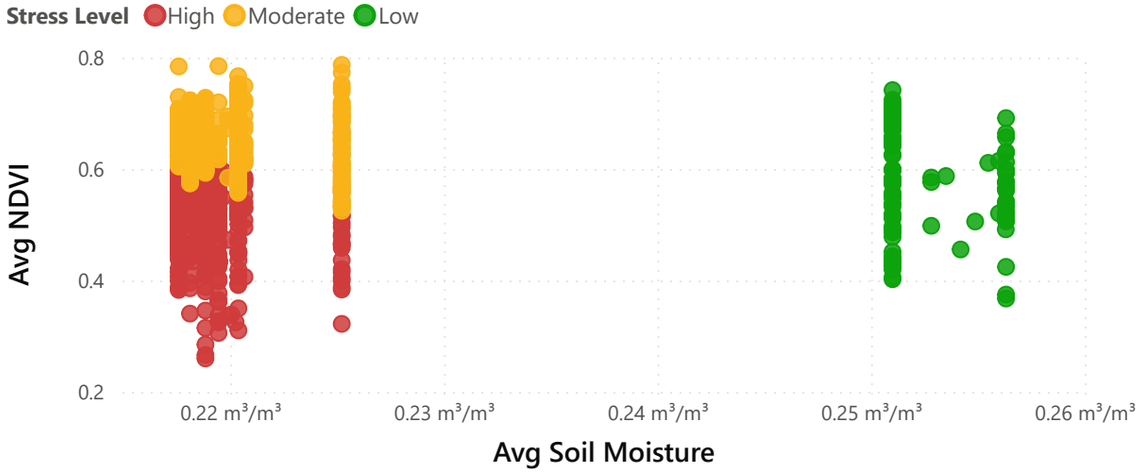
Taluk

All

Average NDVI (crop vigour) by village



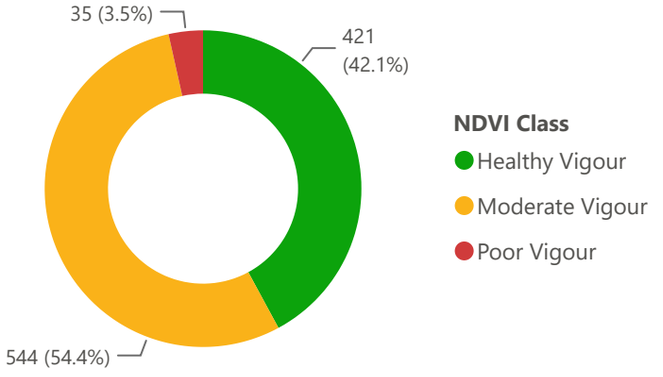
Soil moisture vs NDVI for every plot (colour = stress level)



Soil and crop indicators by taluk and village (expand + to see villages)

Taluk	Farm Count	Avg Soil Moisture	Avg Depletion %	Avg NDVI	Avg LAI	Avg Soil pH	Avg Organic Carbon	Avg Water Stress Scc
⊞ Krishnarajpet	207	0.235 m³/m³	40.8%	0.593	0.58	6.36	10.2	4
⊞ Maddur	102	0.220 m³/m³	50.1%	0.556	0.62	6.65	10.0	6
⊞ Malavalli	204	0.223 m³/m³	48.4%	0.571	0.61	6.44	10.8	5
⊞ Mandya	189	0.219 m³/m³	50.7%	0.569	0.67	6.25	10.1	6
⊞ Pandavpura	205	0.218 m³/m³	51.5%	0.578	0.55	6.29	11.5	6
⊞ Shrirangapattana	93	0.220 m³/m³	49.8%	0.621	0.59	6.38	10.0	5
Total	**1,000**	**0.223 m³/m³**	**48.2%**	**0.580**	**0.60**	**6.37**	**10.5**	**5**

Plots by crop vigour (NDVI class)



Weather and Crop Water Balance

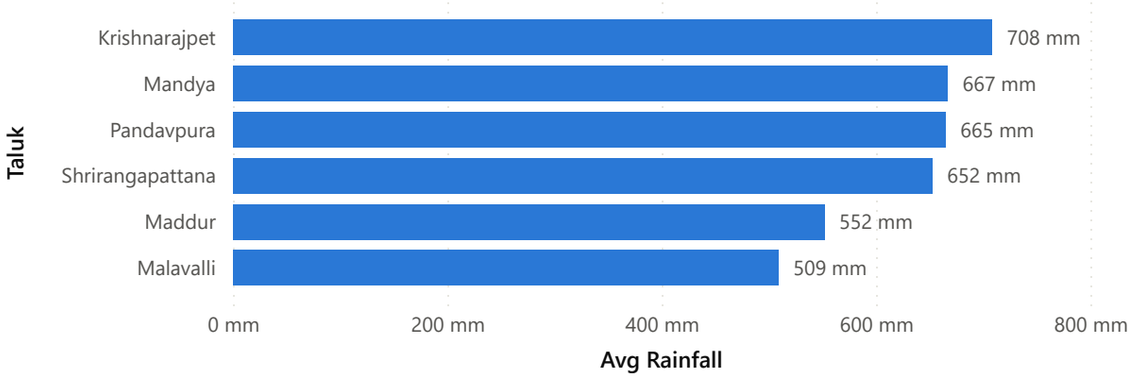
Rainfall, temperature, humidity and crop water use (ETc) across the district

Taluk

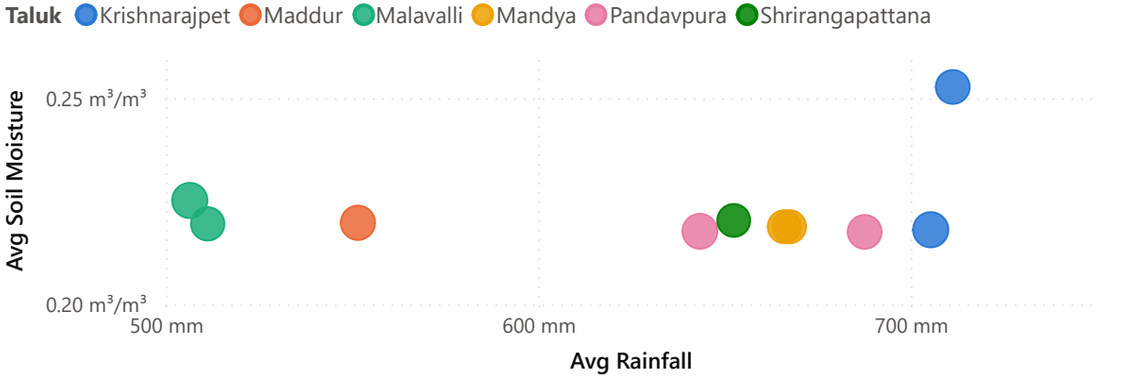
All

Avg seasonal rainfall	Avg temperature	Avg relative humidity	Avg vapour pressure deficit	Sugarcane water use (ETc)
630 mm	23.3 °C	78.9%	0.60 kPa	5.60 mm/day

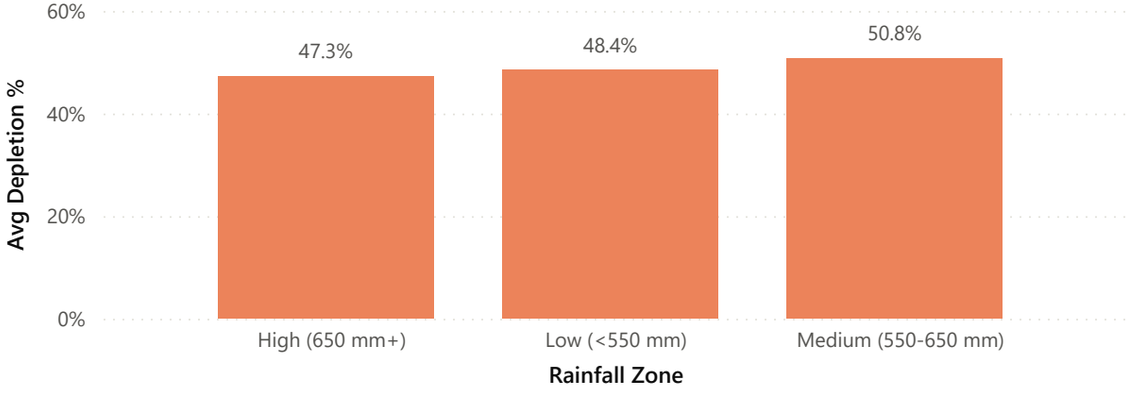
Average seasonal rainfall by taluk



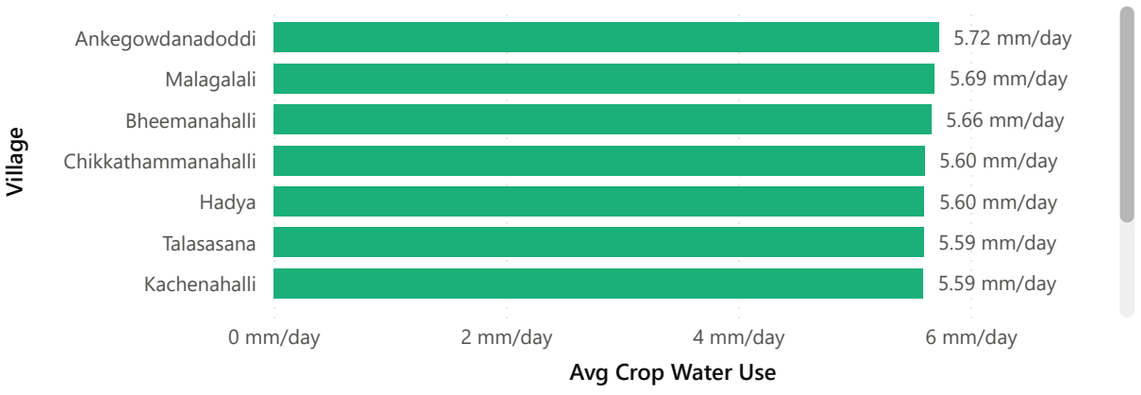
Rainfall vs soil moisture by village (bubble = plots)



Average soil-water depletion by rainfall zone



Sugarcane water use (ETc, mm/day) by village



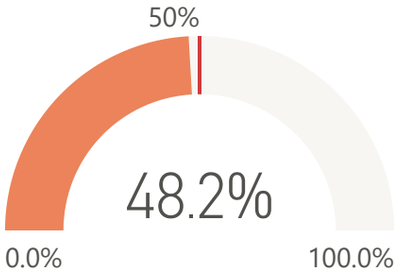
Irrigation Advisory and Decision Support

Plot-level schedule sorted by urgency. Right-click a plot and choose Drill through > Farm Details.

Taluk

All

Avg soil-water depletion vs 50% trigger



Pump hours needed today

12,289

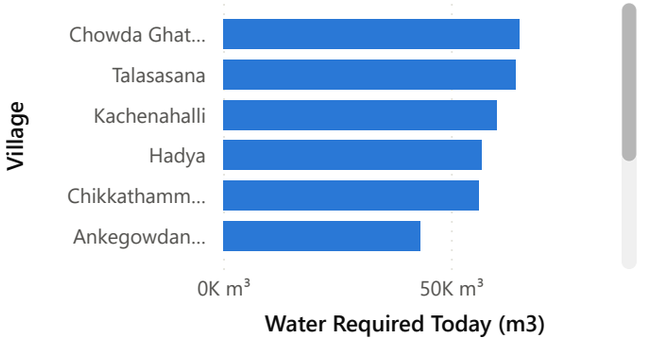
Electricity needed today

45,839 k...

Yield loss risk if 3-day delay

38.5%

Water required today by village (m³)



Advisory summary for the selection

Advisory

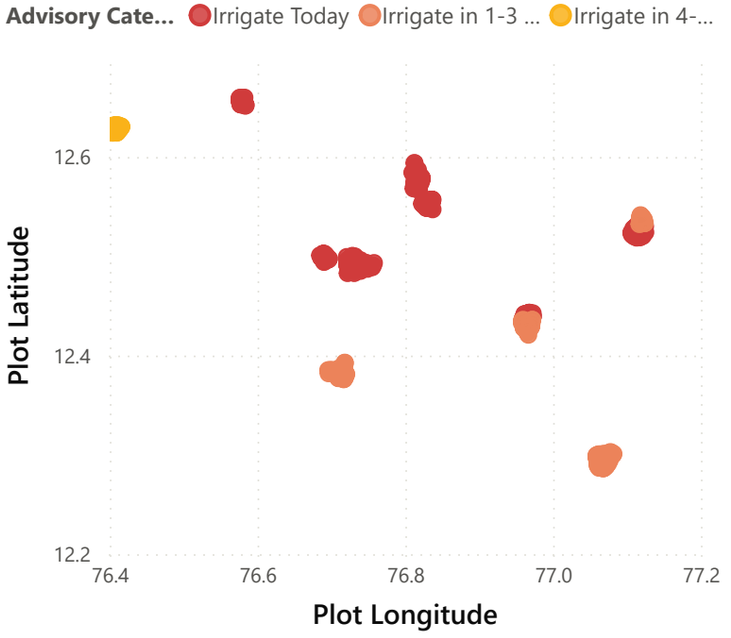
614 of 1,000 plots in the selected area should be irrigated today, needing 368,680 m³ and 12,289 pump hours. Average depletion is 48% against a 50% trigger. Schedule pumps across the 7-hour

Plot-wise irrigation schedule (most urgent first)

Farm_ID	Village	Soil moisture	Depletion	Stress	Days to irrigate	Pump h	Power slots	Advisory
MM-MD-0313	Ankegowdanadoddi	0.219	50.3%	76.5	0.0	19.7	3	Irrigate Today
MM-MD-0566	Ankegowdanadoddi	0.219	50.3%	76.5	0.0	19.7	3	Irrigate Today
MM-MD-0628	Ankegowdanadoddi	0.219	50.3%	76.5	0.0	19.7	3	Irrigate Today
MM-MD-0773	Ankegowdanadoddi	0.219	50.3%	76.5	0.0	19.7	3	Irrigate Today
MM-MD-0530	Ankegowdanadoddi	0.219	50.3%	75.8	0.0	19.7	3	Irrigate Today
MM-MD-0063	Ankegowdanadoddi	0.219	50.3%	75.5	0.0	19.7	3	Irrigate Today
MM-MD-0149	Hadya	0.219	50.7%	75.1	0.0	19.9	3	Irrigate Today
MM-MD-0210	Hadya	0.219	50.7%	75.1	0.0	19.9	3	Irrigate Today
MM-MD-0647	Hadya	0.219	50.7%	75.1	0.0	19.9	3	Irrigate Today
MM-MD-0738	Chikkathammanahalli	0.219	50.7%	75.1	0.0	19.9	3	Irrigate Today
MM-MD-0780	Hadya	0.219	50.7%	75.1	0.0	19.9	3	Irrigate Today
MM-MD-0220	Bheemanahalli	0.220	50.0%	75.0	0.0	19.6	3	Irrigate in 1-3 Days
MM-MD-0888	Ankegowdanadoddi	0.219	50.3%	74.8	0.0	19.7	3	Irrigate Today
MM-MD-0956	Bheemanahalli	0.220	49.8%	74.8	0.0	19.6	3	Irrigate in 1-3 Days
MM-MD-0782	Bheemanahalli	0.220	49.8%	74.7	0.0	19.5	3	Irrigate in 1-3 Days
MM-MD-0997	Bheemanahalli	0.220	49.8%	74.7	0.0	19.5	3	Irrigate in 1-3 Days

Total

Plot map (centroids) by advisory



Scenario Planner and AI Visuals

Move the sliders: expected rain, pump size and irrigation method (efficiency). Ask questions in plain English below.

Rain expected in next 3 da...

Pump flow (m³/h)

Irrigation efficiency (60% f...

Plots to irrigate

614

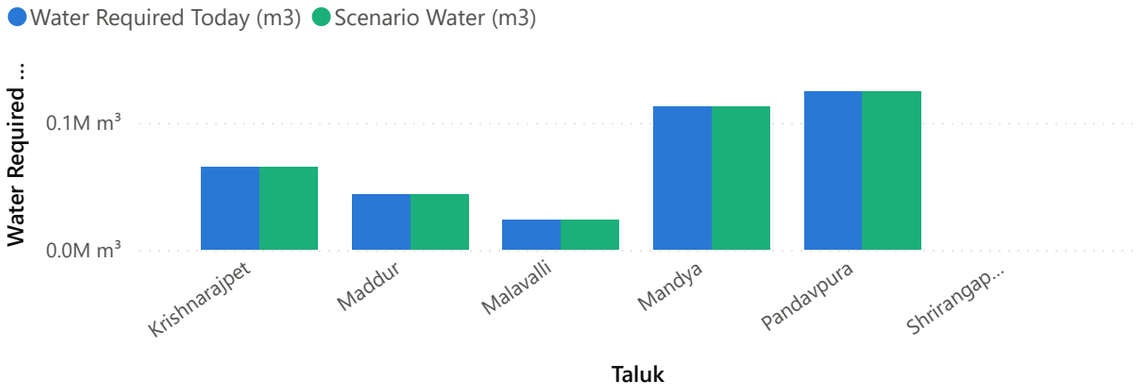
Water needed

3,68,680 m³

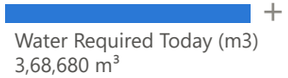
Pump hours

12,289

Water needed: today vs scenario, by taluk (m³)



Decomposition tree: where today's irrigation water goes



Q&A will be retiring in February 2027. Remove Q&A

Ask a question about your data

Try one of these to get started

top village locations by total area (ha)

what is the total area (ha) by village location

Show all suggestions

Farm Details

Drill-through page: one plot's sensor readings, water balance and advisory

Soil moisture	Depletion	Stress score	NDVI	Days to next irrigation
0.223 m^3/m^3	48.2%	56.8	0.580	0.4

Advisory for this plot

Advisory
614 of 1,000 plots in the selected area should be irrigated today, needing 368,680 m^3 and 12,289 pump hours. Average depletion is 48% against a 50% trigger. Schedule pumps across the 7-hour power windows and prioritise plots marked High stress.

Plot attributes

Farm_ID	Taluk	Village	Farm_Area_ha	LAI	Soil_pH	Organic_Carbon	Rainfall_mm	Temperature_C	Relative_Humidity	Latitude	Longitude
MM-MD-0001	Krishnarajpet	Chowda Ghatta	0.736	0.58	6.20	10.0	705	22.2	81.0	12.65639	76.57740
MM-MD-0002	Krishnarajpet	Chowda Ghatta	0.736	0.51	6.40	10.0	705	22.2	81.0	12.65387	76.57776
MM-MD-0003	Krishnarajpet	Laxmipura	0.736	0.60	6.40	10.0	711	22.5	82.7	12.63079	76.41381
MM-MD-0004	Krishnarajpet	Chowda Ghatta	0.736	0.53	6.39	10.0	705	22.2	81.0	12.65854	76.58126
MM-MD-0005	Maddur	Ankegowdanadoddi	0.736	0.61	6.80	10.0	552	24.2	76.2	12.53314	77.12429

Water balance and fertigation

Root Zone Deficit (mm)	Gross Irrigation (mm)	Irrigation Volume (m3)	Pump Hours	Power Slots (7 h)	Energy (kWh)	Yield Loss Risk if Delayed 3 Days	Fertigation Advice
26.2	43.7	321	10.7	2	40	0.0%	Normal vigour: continue the scheduled NPK fertigation.
26.2	43.7	321	10.7	2	40	0.0%	Low vigour: test leaf N, give a split nitrogen dose through drip/furrow wi
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26.2	43.7	321	10.7	2	40	0.0%	Normal vigour: continue the scheduled NPK fertigation.
26.2	43.7	321	10.7	2	40	0.0%	Normal vigour: continue the scheduled NPK fertigation.
26.2	43.7	321	10.7	2	40	0.0%	Normal vigour: continue the scheduled NPK fertigation.
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